AI Content & Marketing Automation Agency

Project Documentation (Version 2) | Track: AI Automation with n8n | Group: ALX5_AIS9_S2 | Company: Skill Dynamix | Round 5 | Supervisor: Ahmed Madeh

GitHub repository: https://github.com/nhahub/NHA-5-247.git

Team: Mustafa Ahmed Mohamed Othman (Team Leader), Omar Nasser Mahmoud, Ziad Mourssy Abdelsamed Oun, Sama Islam Gomail, Andrew Ibrahim Aziz

1. Project Overview

1.1 Summary

The AI Content & Marketing Automation Agency is a set of connected n8n workflows that act like a small digital marketing agency. A client submits one brief (company, industry, tone, audience, goal, platforms) and can add products and weekly instructions at any time. The system then produces marketing content, creates visuals, sends the draft for human approval, publishes it, runs personalized email campaigns, scores leads, and delivers a weekly AI-written performance report.

1.2 Problem Statement

Small businesses cannot afford a full marketing team. Writing posts, designing images, posting on time, following up with leads, and reporting results are repetitive tasks that cost hours every week and are often done inconsistently.

1.3 Objectives

- Turn a single client brief into ready-to-publish content for several platforms.
- Keep the brand voice consistent for each client.
- Let clients add new products and weekly focus without rebuilding the brief.
- Keep a human in the loop: nothing is published without approval.
- Capture leads and follow up with them automatically.
- Give each client a weekly report without manual work.

1.4 Tools and Technologies

Area	Tool
Automation platform	n8n
LLM	Groq (OpenAI or Gemini as alternatives)
Image generation	Image generation API (DALL·E or Stable Diffusion)
Database	Google Sheets
Email	Gmail
Approval and alerts	Telegram Bot
Publishing	Telegram channel, Facebook or LinkedIn where API access allows
Source control	GitHub

2. Project Planning

2.1 Scope

In scope: client intake form, weekly check-in, product onboarding, content generation, quality check, image creation, approval, publishing or scheduling, lead capture, AI email campaigns, lead scoring, weekly report, error alerts.

Out of scope: paid advertising management, a custom web dashboard, payment handling, and the optional WhatsApp status chatbot (only if time remains).

2.2 Team Roles and Responsibilities

Member	Role	Responsibilities
Mustafa	Project Lead and Orchestrator	Intake form, weekly check-in scheduler, central Google Sheet, router between workflows, error alerts, final integration, GitHub repository

Member	Role	Responsibilities
Omar	Content Writer Agent	Blog posts, captions, ad copy, SEO keywords, hashtags, brand-voice prompt, product launch content, saving drafts
Ziad	Visual Creator and Publishing	Image prompts and generation, Approve/Reject flow on Telegram, publishing and scheduling, status updates
Sama	Email Marketing and Lead Generation	Lead capture, personalized emails, lead scoring, follow-up sequence, unsubscribe handling
Andrew	Analytics, QA and Presentation	Weekly report, quality-check agent, test cases, slides, demo video, README

2.3 Work Breakdown and Timeline

Phase	When	Work	Owner
1. Setup	Fri Oct 30, evening	Shared Sheet and column names, GitHub repo, shared credentials, status values	Mustafa
2. Build	Fri night to Sat midday	Each member builds and tests their own workflow	All
3. Integration	Sat afternoon	Connect workflows, end-to-end test with one test client	Mustafa
4. Documentation and slides	Sat afternoon	Finalize documentation, build slides and PDF	Andrew and all
5. Demo and submission	Sat evening	Record video, push JSON exports to GitHub, submit three links	All

Deadline: Saturday, October 31, 2026, 11:59 PM.

2.4 Deliverables

1. Documentation link (this document).

2. GitHub link: https://github.com/nhahub/NHA-5-247.git (exported workflow JSON files and a README).
3. Presentation link: video demonstration and PDF slides.

2.5 Shared Agreements

- One shared Google Sheet with tabs `Clients`, `Products`, `Content`, `Leads`, `Reports`, `Logs`.
- Standard status values: `New → Draft → Approved → Scheduled → Published` (plus `Rejected`).
- One workflow per member, exported as JSON and stored in the shared repository.
- Shared credentials agreed in advance (one Groq key, one Gmail account, one Telegram bot).

2.6 Risks and Mitigation

Risk	Impact	Mitigation
Social platform API approval (for example Meta) takes too long	Cannot publish live	Publish to a Telegram channel for the demo
LLM rate limits or downtime	Workflow stops	Retry on fail, fallback to a second model
Inconsistent column names between members	Workflows break	Fix the column list on day one (Section 4)
Client does not answer the weekly check-in	Workflow stalls	Wait node with timeout, then default settings from the stored brief
Short deadline	Unfinished features	Cut optional features, test with one client
API keys exposed in GitHub	Security issue	Export workflows without credentials, use n8n credential store

3. Stakeholder Analysis

3.1 Stakeholders

Stakeholder	Role	Interest	Influence	Needs
Client (business owner)	Submits the brief, adds products, approves content, receives reports	High	High	Quality content, brand consistency, saved time, clear reports
Agency team (the five members)	Builds and operates the system	High	High	A working, demonstrable, maintainable system
Supervisor (Ahmed Madeh) and training program	Evaluates the project	High	High	Complete deliverables, clear documentation, working demo
Leads and end customers	Receive marketing emails and see posts	Medium	Medium	Relevant messages, an easy way to unsubscribe
Platform providers (Google, Telegram, Meta, LLM provider)	Supply the APIs	Low	Medium	Fair use within their terms and rate limits

3.2 Power and Interest Grid

- **Manage closely (high power, high interest):** client, agency team, supervisor.
- **Keep informed (low power, high interest):** leads and end customers.
- **Keep satisfied (high power, low interest):** platform providers, through compliance with API terms.

3.3 Stakeholder Requirements Summary

- The client must approve every post before it goes live.
- Emails must include an unsubscribe option.
- Each client's data must stay under their own `client_id`.

- Failures must send an alert so nothing fails silently.

4. Database Design

4.1 Overview

Google Sheets is used as a lightweight relational database. Every table links to `Clients` through `client_id`.

4.2 Entity Relationships

- One **Client** has many **Products**.
- One **Client** has many **Content** items.
- One **Client** has many **Leads**.
- One **Client** has many weekly **Reports**.
- Workflow failures are written to **Logs**.

4.3 Tab: Clients (stable brief)

Column	Description
client_id	Unique ID (primary key)
company_name	Client company
industry	Business sector
tone	Brand voice (for example friendly, formal)
audience	Target audience
goal	Campaign goal
platforms	Selected platforms
contact_email	Where reports and alerts are sent
telegram_chat_id	Chat for approvals
weekly_focus	This week's focus from the check-in form
weekly_offer	This week's offer or news

Column	Description
posts_per_week	Number of posts requested
avoid_notes	Things to avoid (from the client and from rejected posts)
last_checkin_at	Last weekly check-in time
brief_updated_at	When the stable brief last changed
status	New, Active
created_at	Timestamp

4.4 Tab: Products

Column	Description
product_id	Unique ID (primary key)
client_id	Link to Clients
name	Product name
description	Short description
benefits	Key benefits or unique selling points
price_or_offer	Price or launch offer
launch_date	Launch date
image_url	Product photo (optional)
is_new	Yes for new launches, changed to No after about 14 days
status	Active or Discontinued
created_at	Timestamp

4.5 Tab: Content

Column	Description
content_id	Unique ID (primary key)
client_id	Link to Clients
product_id	Product featured in the post, if any
week_start	Week the post belongs to
platform	LinkedIn, Facebook, Instagram, Telegram, Blog
content_type	Post, caption, blog, ad copy
body	Generated text
hashtags	Suggested hashtags
seo_keywords	Suggested keywords
image_prompt	Prompt used for the image
image_url	Generated image link
qa_status	Passed or Failed quality check
status	Draft, Approved, Rejected, Scheduled, Published
scheduled_at	Planned publish time
published_at	Actual publish time
post_url	Link to the live post
created_at	Timestamp

4.6 Tab: Leads

Column	Description
lead_id	Unique ID (primary key)
client_id	Link to Clients
name	Lead name

Column	Description
email	Lead email
source	Form, webhook, or other source
message	Lead's message or interest
score	Hot, Warm, Cold
score_reason	Short AI explanation
email_stage	Welcome, Follow-up 1, Follow-up 2, Newsletter
last_contacted	Timestamp
unsubscribed	Yes or No
created_at	Timestamp

4.7 Tab: Reports

Column	Description
report_id	Unique ID (primary key)
client_id	Link to Clients
week_start, week_end	Reporting period
posts_published	Number of posts
leads_captured	Number of new leads
emails_sent	Number of emails
hot_leads	Number of hot leads
ai_summary	AI-written summary
sent_to	Delivery address or chat
created_at	Timestamp

4.8 Tab: Logs

Column	Description
log_id	Unique ID
workflow_name	Which workflow failed
error_message	Error text
client_id	Affected client, if known
created_at	Timestamp

5. System Architecture and Workflows

5.1 High-Level Flow

1. The client fills the **brief form** once (Workflow 1) and the row is saved to `Clients`.
2. Every week, a scheduler sends each active client a short **check-in form** (Section 5.3).
3. The client can add a **new product** at any time (Section 5.4).
4. The **Content Writer** builds the prompt from the stable brief, this week's input, active products, and recent posts, then saves drafts as `Draft` (Workflow 2).
5. The **Quality Check** agent reviews grammar, tone, and banned words.
6. The **Visual Creator** generates an image and sends the draft to Telegram for approval (Workflow 3).
7. After approval, the post is published or scheduled across the week and the status is updated.
8. New leads enter through the lead form and receive AI emails (Workflow 4).
9. At the end of the week, the report workflow produces the client report (Workflow 5).
10. Any workflow failure triggers the global error alert.

5.2 Workflow Descriptions

#	Workflow	Owner	Trigger	Main nodes	Output
1	Client Intake, Weekly Check-in and Orchestrator	Mustafa	Form Trigger and weekly Schedule Trigger	Form Trigger, Schedule Trigger, Google Sheets, Loop, Wait, Execute Workflow, Error Trigger, Telegram/Gmail	Saved client, weekly input, started pipeline, error alerts
2	Content Writer Agent	Omar	Called by Workflow 1 or the product form	Google Sheets, Code (prompt builder), LLM chain (Groq) with structured output, Google Sheets	Drafts in `Content`
3	Visual Creator and Publishing	Ziad	Called after Draft	LLM, image API, Telegram (send and wait for approval), IF/Switch, publish node, Google Sheets	Approved and published posts
4	Email Marketing and Leads	Sama	Lead form or webhook	Form/Webhook, Google Sheets, LLM, Gmail, Wait, IF	Scored leads and sent emails
5	Analytics, QA and Reporting	Andrew	Weekly schedule, plus QA called before approval	Schedule Trigger, Google Sheets, Code (counts), LLM, Gmail/Telegram	Weekly client report

5.3 Weekly Input Flow

The client does not repeat the whole brief each week. A **Schedule Trigger** (for example Sunday 9 AM) reads all clients with `status = Active` and loops over them one at a time:

1. Telegram or Gmail sends: "What should we focus on this week?" with a link to a short form (client ID pre-filled).
2. The form asks for: this week's focus, offer or news (optional), number of posts, and things to avoid (optional). It takes about 30 seconds.
3. A **Wait** node holds for the answer (for example 24 hours). If no answer arrives, the workflow continues with default settings from the stored brief, so it never stalls.
4. The answers are saved in `weekly_focus`, `weekly_offer`, `posts_per_week`, and `avoid_notes`, and the Content Writer is started.

5.4 New Product Flow

The brief describes the business, not each product, so products have their own tab and form.

1. The client submits an **Add product** form at any time (name, description, benefits, price or offer, launch date, optional photo).
2. The workflow saves a row in `Products` with `is_new = Yes` and starts a **launch campaign** immediately.
3. The Content Writer creates a launch set: an announcement post, a benefits post, an offer post, and a teaser email.
4. In the following weeks, new products are featured in at least two posts. After about 14 days `is_new` becomes `No` and the product joins the normal rotation. Products marked `Discontinued` are never mentioned.

5.5 How the Brief Stays Up to Date

The prompt is built from three layers:

Layer	Source	Changes
1. Stable brief	`Clients` row	Only when the client updates it
2. Weekly overlay	Weekly form and `Products` tab	Every week or when a product is added
3. History	Last posts in `Content`, rejected posts, `avoid_notes`	Automatically

The AI never edits the stable brief by itself. Only the client's form can change it, which keeps the brand voice safe.

5.6 Quality Check Rules (Workflow 5)

- Grammar and spelling are acceptable.
- Tone matches the client's `tone` field.
- No banned words or unsupported claims.
- If the check fails, the content is returned to the writer for one revision.

5.7 Error Handling

A shared Error Trigger workflow writes the failure to `Logs` and sends a Telegram or Gmail alert to the team lead with the workflow name and error text.

5.8 Security Notes

- Credentials stay in the n8n credential store and are removed from exported JSON files.
- Only the minimum lead data needed is stored.
- Unsubscribed leads are never emailed again.

6. UI/UX Design

The system has no separate website. The interfaces are the n8n forms, Telegram messages, emails, and Google Sheets views. They follow three principles: few steps, clear wording, and one obvious action per screen.

6.1 Screen 1: Client Brief Form (n8n Form)

Field	Type	Notes
Company name	Text	Required
Industry	Dropdown	For example retail, food, tech, education
Brand tone	Dropdown	Friendly, professional, playful, luxury
Target audience	Text	Required
Campaign goal	Dropdown	Awareness, sales, leads
Platforms	Multi-select	LinkedIn, Facebook, Instagram, Telegram
Contact email	Email	Required for reports

Result page: "Thanks! Your content is being prepared."

6.2 Screen 2: Weekly Check-in Form

Four short fields: this week's focus (dropdown), offer or news (optional text), posts wanted (3, 5, or 7), and things to avoid (optional text).

6.3 Screen 3: Add Product Form

Product name, short description, key benefits, price or offer, launch date, optional photo, and priority (new launch or regular).

6.4 Screen 4: Telegram Approval Message

The client sees the image, post text, hashtags, and platform, followed by two buttons:

- **Approve** - the post is published or scheduled.
- **Reject** - the status becomes `Rejected` and the client may add a reason, which is stored in `avoid_notes`.

6.5 Screen 5: Lead Form

Name, email, and a message. After submission, the lead receives a welcome email within minutes.

6.6 Screen 6: Email Templates

- **Welcome email:** personalized greeting, one sentence about how the client can help, one call-to-action button, unsubscribe link.
- **Follow-up email:** references the lead's message, one clear next step.
- **Newsletter:** short intro and two to three highlights, including new products.

6.7 Screen 7: Weekly Report Email

Subject: "Your weekly marketing report". A three-line AI summary, a small table with posts published, leads captured, emails sent, and hot leads, and a recommended action for next week.

6.8 Screen 8: Google Sheet Dashboard

Colour-coded status column (grey Draft, blue Approved, yellow Scheduled, green Published, red Rejected), frozen header rows, and filters by `client_id`.

6.9 UX Principles Applied

- Dropdowns replace free text wherever possible to reduce errors.
- Every automated message ends with a clear next step.
- Human approval is always required before publishing.

- Mobile-friendly: forms and approvals work from a phone.

Screenshots of each screen should be added here before submission.

7. Testing

ID	Test case	Steps	Expected result
T1	Brief submission	Submit the form with valid data	New row in `Clients`, pipeline starts
T2	Content generation	Run Workflow 2 for a test client	Drafts saved with status `Draft` and the correct tone
T3	Quality check	Include a banned word in a draft	Check fails and content is revised
T4	Approval	Press Approve in Telegram	Status becomes `Approved`, then `Scheduled` or `Published`
T5	Rejection	Press Reject with a reason	Status becomes `Rejected`, reason saved in `avoid_notes`
T6	Lead capture and scoring	Submit a lead form	Lead saved with a Hot, Warm, or Cold score
T7	Email follow-up	Wait for the follow-up step	Follow-up sent, `email_stage` updated
T8	Unsubscribe	Mark a lead as unsubscribed	No further emails are sent
T9	Weekly report	Run the schedule manually	Report row created and email sent
T10	Error alert	Force a workflow to fail	Row in `Logs` and alert received
T11	Weekly check-in	Run the weekly schedule and submit the form	Weekly fields saved and posts reflect the focus
T12	Check-in timeout	Run the schedule and do not answer	Workflow continues with defaults from the brief
T13	New product	Submit the Add product form	Row in `Products` and launch posts mention the product

8. Conclusion and Future Work

The project shows that a full content-to-report marketing cycle can be automated with n8n, LLMs, and simple tools, while keeping a human approval step. The client gives a brief once, then adds products and weekly input in under a minute. Possible future work:

- A WhatsApp or Telegram chatbot where clients ask for their campaign status.
- Direct publishing to more platforms after API approval.
- Multi-language content generation (for example Arabic and English).
- A web dashboard with live analytics.

9. Links

- GitHub repository: https://github.com/nhahub/NHA-5-247.git
- Demo video: *(add link)*
- Presentation PDF: *(add link)*